FRACTIONS - SHORT REVISION NOTES

Evaluate Level: Make Judgments and Justify Decisions

1. Verifying Whether an Answer Is Correct

Problem	Student's Answer	Verification	Judgment
1/2 + 1/3	2/5	Find LCM: 1/2+1/3 = 3/6+2/6 = 5/6	INCORRECT ✗
1/2 + 1/3	5/6	3/6+2/6 = 5/6	CORRECT ✓
3/4 - 1/2	1/4	3/4-1/2 = 3/4-2/4 = 1/4	CORRECT ✓
3/4 x 2/3	1/2	3/4 x 2/3 = 6/12 = 1/2	CORRECT ✓
1/2 ÷ 1/3	1/6	1/2 x 3/1 = 3/2 = 1 1/2	INCORRECT ✗
1/2 ÷ 1/3	1 1/2	1/2 x 3/1 = 1 1/2	CORRECT ✓

Verification Method:

- Perform the calculation yourself
- Compare with the student's answer
- If they match → CORRECT
- If they don't match → INCORRECT, explain WHY it's wrong

2. Judging the Best Method

Problem	Method A	Method B	Which is Better?
1/2 + 1/3	Find LCM = 6	Use calculator	Method A - shows understanding ✓
3/4 x 2/3	Multiply: 3/4 x 2/3 = 6/12 = 1/2	Decimals: 0.75x0.666=0.5	Both - same answer
5/8 ÷ 7/8	Keep, Change, Flip: 5/8 x 8/7 = 40/56 = 5/7	Reciprocal concept	Method A - teaches concept ✓
1/2 + 1/4 + 1/3	LCM = 12	Add two at a time	LCM - more efficient

Evaluation Criteria:

- Which method is faster?
- Which method shows better understanding?
- Which method is less likely to cause errors?

3. Evaluating the Reasonableness of Answers

Problem	Answer	Reasonable?	Justification
1/2 x 100	50	YES ✓	Half of 100 is 50
1/3 x 90	30	YES ✓	90 ÷ 3 = 30
3/4 x 20	25	NO ✗	3/4 of 20 should be 15 (20÷4x3=15)
1/2 ÷ 1/3	1/2	NO ✗	Dividing by a fraction gives a larger answer; should be 1 1/2
1/3 ÷ 1/2	2/3	YES ✓	1/3 x 2/1 = 2/3

Reasonableness Checks:

If multiplying proper fractions: result should be SMALLER than each factor ✓ (if larger → wrong ✗)
If dividing by a proper fraction: result should be LARGER than the first fraction ✓ (if smaller → wrong ✗)
If finding a fraction of a whole: result should be LESS than the whole ✓ (if greater than whole → wrong ✗)

4. Judging the Simplest Form

Fraction	Simplified?	Judgment	Justification
2/4	NO ✗	Can divide by 2 → 1/2	
3/6	NO ✗	Can divide by 3 → 1/2	
4/8	NO ✗	Can divide by 4 → 1/2	
1/2	YES ✓	No common factor	
3/4	YES ✓	No common factor	
6/8	NO ✗	Can divide by 2 → 3/4	
4/9	YES ✓	No common factor	
12/16	NO ✗	Can divide by 4 → 3/4	

Simplification Judgment:

Fraction	HCF	Divided by	Simplified	Judged?
6/9	3	÷3	2/3	✓ Correct
8/12	4	÷4	2/3	✓ Correct
10/25	5	÷5	2/5	✓ Correct
21/28	7	÷7	3/4	✓ Correct
35/40	5	÷5	7/8	✓ Correct

5. Evaluating the LCM Choice

Fractions	LCM Choices	Correct LCM	Why?
1/2 + 1/3	6 or 12	6	Least Common Multiple
3/4 + 4/5	20 or 40	20	Least Common Multiple
2/3 + 3/4 + 1/2	12 or 24	12	Least Common Multiple
5/6 + 7/8	24 or 48	24	Least Common Multiple

Judgment Criteria:

LCM is the SMALLEST number both denominators divide into. Using a larger common denominator still works but is LESS EFFICIENT.
1/2+1/3 using 12: 6/12+4/12 = 10/12 = 5/6 → works but not efficient.
1/2+1/3 using 6: 3/6+2/6 = 5/6 → BETTER.

6. Judging Student Mistakes

Student's Work	Analysis	Judgment	Correction
1/2 + 1/3 = 2/5	Added numerators and denominators	INCORRECT ✗	Need common denominator: 5/6
3/4 - 1/2 = 1/4	Worked correctly	CORRECT ✓	3/4 - 2/4 = 1/4
1/2 x 1/3 = 1/6	Worked correctly	CORRECT ✓	1x1=1, 2x3=6
3/4 ÷ 1/2 = 3/8	Multiplied instead of dividing?	INCORRECT ✗	Should be 3/4 x 2/1 = 3/2 = 1 1/2
2/3 of 3/4 = 1/2	Worked correctly	CORRECT ✓	2/3 x 3/4 = 6/12 = 1/2
12/16 = 3/4	Simplified correctly	CORRECT ✓	÷4 = 3/4

Common Mistake Analysis Table:

Mistake Type	Example	Why Wrong	Correct Method
Adding denominators	1/2+1/3 = 2/5	Can't add different sizes	Find LCM: 3/6+2/6=5/6
Wrong reciprocal	3/4÷1/2 = 3/4x1/2=3/8	Should flip divisor	3/4x2/1=1 1/2
Not simplifying	6/8 = not simplified	Has common factor 2	6/8 = 3/4
Confusing "of"	1/2 of 20 = 20 1/2	"of" means x	1/2x20=10

7. Evaluating Operation Choice

Problem Phrase	Operation	Why This Operation?
"Find 1/2 of 100"	Multiplication (x)	"of" means multiply
"What is left?"	Subtraction (-)	Removing a part from the whole
"How much in total?"	Addition (+)	Combining parts together
"How many times does it fit?"	Division (÷)	Comparing how many of one fits in another
"What fraction is this?"	Division (÷)	Part ÷ Whole

Word Problem Operation Judgment:

Word Problem	Chosen Operation	Correct?	Justification
"Find 2/3 of 30"	x	YES	"of" = multiplication
"Spent 1/4, spent 1/3"	+	YES	Add to find total spent
"What fraction is left?"	-	YES	Whole - used = remaining
"How many 1/5s in 1/2?"	÷	YES	1/2 ÷ 1/5 = 2 1/2
"Divide 10 by 1/2"	÷	YES	10 ÷ 1/2 = 20

8. Evaluating Equivalent Fractions

Claim	Verification	Judgment
1/2 = 2/4	1x4=4, 2x2=4	TRUE ✓
1/3 = 2/6	1x6=6, 3x2=6	TRUE ✓
3/4 = 6/8	3x8=24, 4x6=24	TRUE ✓
2/5 = 4/10	2x10=20, 5x4=20	TRUE ✓
4/5 = 15/20	4x20=80, 5x15=75	FALSE ✗
5/6 = 10/12	5x12=60, 6x10=60	TRUE ✓
7/8 = 14/16	7x16=112, 8x14=112	TRUE ✓

Verification Method: Cross multiply: a×d = b×c

9. Comparing Two Solutions

Problem	Solution A	Solution B	Evaluation
1/2 + 1/3	3/6 + 2/6 = 5/6	6/12 + 4/12 = 10/12 = 5/6	Both correct, A is simpler
3/4 x 4/5	3/4 x 4/5 = 12/20 = 3/5	0.75 x 0.8 = 0.6 = 3/5	Both correct, A shows understanding
1/2 ÷ 1/3	1/2 x 3/1 = 1 1/2	0.5 ÷ 0.333 = 1.5	Both correct, A is more educational

Evaluation Criteria:

- Which solution is clearer?
- Which solution shows better understanding?
- Which solution is more efficient?
- Which solution has fewer errors?

10. Evaluating "Of" Problems

Problem	Answer	Correct?	Justification
1/2 of 64	32	YES ✓	64 ÷ 2 = 32
1/2 of 64	128	NO ✗	That's 64 x 2, not 1/2 x 64
1/3 of 6	18	NO ✗	6 ÷ 3 = 2, not 18
1/3 of 6	2	YES ✓	6 ÷ 3 = 2
3/4 of 20	15	YES ✓	20 ÷ 4 x 3 = 15
3/5 of 40	24	YES ✓	40 ÷ 5 x 3 = 24

Common Mistakes in "Of" Problems:

Error	Example	Why Wrong	Correct
Adding instead of multiplying	1/2 of 20 = 20 1/2	"of" means x	1/2 x 20 = 10
Multiplying numerator only	2/3 of 15 = 30	Need to divide first	15 ÷ 3 x 2 = 10
Wrong order (this is actually correct!)	1/2 of 64 = 64 ÷ 2 ✓	-	✓ Correct
Confusing numerator/denominator (correct!)	3/4 of 20 = 15 ✓	-	✓ Correct

11. Evaluating Reciprocals

Number	Student's Reciprocal	Correct?	Why?
2/3	3/2	YES ✓	Flipped correctly
3/4	4/3	YES ✓	Flipped correctly
5	5	NO ✗	Should be 1/5 (5 = 5/1 → reciprocal = 1/5)
5	1/5	YES ✓	Correct
2 1/2 (5/2)	2/5	YES ✓	Convert to improper first
1 3/4 (7/4)	4/7	YES ✓	Convert to improper first
1/5	5	YES ✓	1/5 → reciprocal = 5/1 = 5

Reciprocal Checklist:

- Is it a proper fraction? → Flip it!
- Is it a whole number? → Write as a fraction (n/1) → Flip!
- Is it a mixed number? → Convert to improper first → Flip!

12. Evaluating BODMAS Application

Expression	Student's Work	Correct Order?	Judgment
2 + 3 x 4	5 x 4 = 20	NO ✗	Should be 2 + 12 = 14
2 + 3 x 4	2 + 12 = 14	YES ✓	Correct order
(2 + 3) x 4	5 x 4 = 20	YES ✓	Brackets first
1/2 x 3 + 1	1/2 x 4 = 2	NO ✗	"of" not present; multiply then add: 1 1/2+1=2 1/2
1/2 x (3 + 1)	1/2 x 4 = 2	YES ✓	Brackets first
6 ÷ 2 x 3	6 ÷ 6 = 1	NO ✗	Division then multiplication, left to right: 3x3=9

BODMAS Priority Evaluation:

Operation	Priority	Example	Correct Order
Brackets	1st	(1/2+1/3)x1/4	5/6 x 1/4 = 5/24
Of	2nd	1/2 x 1/3 + 1/4	1/6 + 1/4 = 5/12
Division	3rd (L to R)	6÷2x3	3x3=9
Multiplication	3rd (L to R)	6÷2x3	3x3=9
Addition	4th (L to R)	5+3-2	8-2=6
Subtraction	4th (L to R)	5+3-2	8-2=6

13. Judging Word Problem Answers

Example 1: Money Problem

Problem	Student Answer	Correct?	Justification
1/4 of income for food, 1/2 for business. What fraction saved?	1/4	YES ✓	1-(1/4+1/2) = 1-3/4 = 1/4
Same problem	3/4	NO ✗	That's what's spent, not saved

Example 2: Land Problem

Problem	Student Answer	Correct?	Justification
Father gives 1/2 to son, 1/3 to daughter. Son donates 1/5 of his portion. Fraction of total land donated?	1/10	YES ✓	1/2 x 1/5 = 1/10
Same problem	3/5	NO ✗	That's total given away

Example 3: Journey Problem

Problem	Student Answer	Correct?	Justification
Walked 1/8, train 2/3. What fraction by bus?	5/24	YES ✓	1-(1/8+2/3) = 1-(3/24+16/24) = 1-19/24 = 5/24
Same problem	19/24	NO ✗	That's what was spent, not remaining

14. Comparing Order of Operations

Expression	Using BODMAS	Ignoring BODMAS	Correct?
2 + 3 x 4	2 + 12 = 14	5 x 4 = 20	BODMAS ✓
1/2 x 3 + 1	1 1/2 + 1 = 2 1/2	1/2 x 4 = 2	BODMAS ✓
6 ÷ 2 x 3	3 x 3 = 9	6 ÷ 6 = 1	BODMAS ✓
1/2 + 1/3 x 3/4	1/2 + 1/4 = 3/4	5/6 x 3/4 = 5/8	BODMAS ✓

Why BODMAS Is Better:

- Ensures everyone gets the same answer
- Prevents ambiguity
- Is the accepted convention worldwide

15. Evaluation Checklist

When verifying an answer

- ■ Did I perform the calculation myself?
- ■ Do I get the same answer?
- ■ If not, where did the student go wrong?
- ■ Is the answer reasonable?

When comparing methods

- ■ Which method is more efficient?
- ■ Which method shows better understanding?
- ■ Which method has fewer steps?
- ■ Which method is less error-prone?

When checking reasonableness

- ■ For multiplication: Is the result smaller than each factor?
- ■ For division: Is the result larger than the first fraction?
- ■ For "of": Is the result less than the whole?
- ■ For simplification: Can it be simplified further?

When evaluating a student's work

- ■ Did they add/subtract correctly?
- ■ Did they find the correct LCM?
- ■ Did they multiply/divide correctly?
- ■ Did they simplify properly?
- ■ Did they use BODMAS correctly?
- ■ Did they show their working?

When judging word problems

- ■ Did they identify the correct operation?
- ■ Did they set up the problem correctly?
- ■ Is the answer reasonable?
- ■ Does the answer make sense in context?
- ■ Are the units correct?

16. Self-Test - Evaluation Questions

No.	Question	Answer & Justification
1	Is 1/2 + 1/3 = 5/6 correct?	YES ✓ 3/6 + 2/6 = 5/6
2	Is 3/4 x 2/3 = 1/2 correct?	YES ✓ 6/12 = 1/2
3	Is 3/4 ÷ 1/2 = 3/8 correct?	NO ✗ Should be 1 1/2
4	Is 2/3 of 3/4 = 1/2 correct?	YES ✓ 2/3 x 3/4 = 6/12 = 1/2
5	Should 6/8 be simplified?	YES ✓ Divide by 2 → 3/4
6	Which is better: LCM=6 or LCM=12 for 1/2+1/3?	LCM=6 is more efficient
7	Is 1/2 of 64 = 32 reasonable?	YES ✓ 64÷2=32
8	Is 1/2 of 64 = 128 reasonable?	NO ✗ That's 64x2
9	Is 5 ÷ 1/5 = 1?	NO ✗ 5 x 5 = 25
10	Is 1/3 of 6 = 2 reasonable?	YES ✓ 6÷3=2

17. Quick Evaluation Guide

■ CORRECT if:

- Denominators match when adding/subtracting
- Fractions are in simplest form
- "of" is replaced by multiplication
- Division uses the reciprocal
- BODMAS order is followed
- Answer is reasonable

■ INCORRECT if:

- Adding numerators and denominators separately
- Not finding a common denominator
- Forgetting to flip when dividing
- Not simplifying fractions
- Ignoring BODMAS
- Answer is unreasonable

■ REASONABLE if:

- Multiplication result is smaller than the factors
- Division result is larger than the first fraction
- "of" result is less than the whole
- Answer makes sense in context

Revision Tip: When evaluating, always ask “Does this make sense?” and “Why is this correct or incorrect?” Justification is the key to true understanding!